1 Manifolds

2 Lie groups

Definition (Lie group)**.** *(G, e, m, i) st G smooth, m smooth.*

Proposition. *The differentials of the maps*

- $dm(e,e) : \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}, \quad (X,Y) \mapsto X+Y$
- $di(e) : \mathfrak{g} \to \mathfrak{g}, \quad X \mapsto -X$

Definition (Adjoints)**.**

$$\begin{aligned} \boldsymbol{Ad}(g) : G &\to G \\ h &\mapsto ghg^{-1} \end{aligned} \qquad \begin{aligned} \mathrm{Ad}(g) : \mathfrak{g} &\to \mathfrak{g} \\ X &\mapsto gXg^{-1} \end{aligned} \qquad \begin{aligned} \mathrm{ad} : \mathfrak{g} &\to \mathrm{End}(\mathfrak{g}) \\ X &\mapsto [X, \cdot] \end{aligned}$$

Proposition. *$\varphi : G \to H$. Then*

- $d\varphi(e) \circ Ad(g) = Ad(\varphi(g)) \circ d\varphi(e)$
- $d\varphi(e)([X,Y]_G) = [d\varphi(e)X, d\varphi(e)Y]_H$

Definition (Lie algebra)**.** *Finite dimensional real (or complex) vector space L and bilinear map st*

- $[X,Y] = -[Y,X]$
- $[X,[Y,Z]] + [Y,[Z,X]] + [Z,[X,Y]] = 0$

Homomorphisms $\varphi : L \to M$ satisfy $\varphi([X,Y]_L) = [\varphi(X), \varphi(Y)]_M$

Proposition. *G top. group, $H \subseteq G$ open subgroup $\implies$ H closed.*

Proposition. *G connected top. group, $U \subseteq G$ nbh of e Then $G = \bigcup_{n \geq 1} U^n$.*

Proposition. *Let G top. group, G^0 conn. comp. of 1.*

- *G^0 closed sbg characteristic (stable under all automorphisms)*
- *G locally conn. $\implies$ G^0 open and contained in every open subgroup*
- *Conn. comp. of G are cosets G/G^0.*

Definition (Vector field)**.** *Assigns $v(p) \in T_p(M)$. Smooth if system of local coord. are smooth functions.*

Definition (Integral curve)**.** *(I, γ) st $\gamma : I \to M$, $\gamma'(t) = c(\gamma(t))$*

Definition. $L_g : G \to G, \quad h \mapsto gh$. *Taking differentials $l_g \coloneqq dL_g(1) : \mathfrak{g} \to T_gG$*

Definition (Left invariant)**.** *$v(g) = l_g(v(1))$*

Proposition. *v left inv. Then, maximal integral curve γ of v with $\gamma(0) = e$ is defined on $\mathbb{R}$ and $\gamma(t+s) = \gamma(t)\gamma(s)$.*

Definition. $\exp_G(X) = \gamma_X(1)$, *with $\gamma'_X(0) = X$. If V finite dimensional, then*

$$\exp_G(X) = \sum_{n=0}^{\infty} \frac{X^n}{n!}$$

Proposition. *Porperties of* exp*:*

- $\boldsymbol{Ad}(x) \circ \exp_G = \exp_G \circ \mathrm{ad}(x)$
- $\boldsymbol{Ad} \circ \exp_G = \exp_{GL(\mathfrak{g})} \circ \mathrm{ad}$
- $d\exp_G(0) = id_{\mathfrak{g}}$
- $\varphi \circ \exp_G = \exp_H \circ d\varphi(1)$
- $\gamma_X(t) = \exp_G(tX)$

Proposition. *We have*

- $\exp_G : \mathfrak{g} \to G$ *is a local diffeomorphism at* 0.
- *Image of* $\exp_G$ *lies in* G^0
- $U \subseteq \mathfrak{g}$ *nbh of* 0*, then* $\exp(U)$ *generates* G^0.
- *If* G *is connected,* $\exp_G$ *is surjective.*
- $f : G \to H$*, then* $df_1(X) = \frac{d}{dt} |_{t=0} f(\exp_G(tX))$

3 Compact topological groups

Definition (Separation axioms)**.** *These are*

- T_0*:* $\forall x \neq y$, $\exists U$ *st* $x \in U$, $y \notin U$ *or* $y \in U$, $x \notin U$
- T_1*:* $\forall x \neq y$, $\exists U$ *st* $x \in U$, $y \notin U$ *and* $\exists V$ *st* $y \in V$, $x \notin V$
- T_2*:* $\forall x \neq y$, $\exists U, V$ *st* $x \in U$, $y \in V$, $U \cap V = \emptyset$
- T_3*:* $\forall x \in X$, $\forall F \subseteq X$ *closed,* $x \notin F$, $\exists U$ *open st* $x \in U$, $F \cap U = \emptyset$
- $T_{3\frac{1}{2}}$*:* $\forall x \in X$, $\forall F \subseteq X$ *closed,* $x \notin F$, $\exists f : X \to [0, 1]$ *st* $f(x) = 0$, $f(F) = \{1\}$
- T_4*:* $\forall F, F' \subseteq X$ *closed,* $F \cap F' = \emptyset$, $\exists U, V$ *st* $F \subseteq U$, $F' \subseteq V$, $U \cap V = \emptyset$

Proposition. *For top. groups,* $T_0 \implies T_{3\frac{1}{2}}$.

Proposition (Urysohn's lemma)**.** T_4 *is equivalent to* $\forall F, F'$ *closed* $\exists f : X \to [0, 1]$ *continuous,* $f(F) = \{0\}$, $f(F') = \{1\}$.

Proposition. T_2 *and compact* $\implies T_4$.

Let S a set, Λ a comm. ring

Definition (Algebraic S-Mod)**.**

- M *S-module over* Λ *if* M *is* Λ*-module and* $a = a_M : S \to \mathrm{End}_\Lambda(M)$
- $N \subseteq M$ *S-submodule if* N *is* Λ*-submodule and* $a(S)(N) \subseteq N$
- $f : M \to N$ *S-morphism if* f *is* Λ*-linear and* $f(a_M(s)m) = a_N(s)f(m)$ *for all* $s \in S$

Definition. *Algebraic definitions:*

- ***Irreducible*** M *has no non-zero proper S-submodules.*
- ***Indecomposable*** M *is not a direct sum of two non-zero S-modules.*
- ***Semisimple*** M *is a direct sum of irreducible S-modules.*

Definition (Length)**.** *Supremum of* $M_1 \subsetneq M_2 \subsetneq \cdots \subsetneq M_n = M$ *of S-submodules of* M.

Proposition. M *semisimple* $\iff$ M *is a sum of irreducible S-modules*

If that happens, then there exists a (non unique) complement for each submodule. If M *is finite length, all are equivalent.*

Proposition. *Let* $f : M \to N$. *If* M *irreducible, then* $f = 0$ *or* f *is injective. If* N *irreducible, then* $f = 0$ *or* f *is surjective.*

Proposition (Schur's lemma)**.** V, W *irreducible fin. dim.* $\mathbb{C}$*-vector spaces. Then*

$$\dim \mathrm{Hom}_S(V, W) = \begin{cases} 1 & V \cong W \\ 0 & \textit{otherwise} \end{cases}$$

Definition. $\hat{S}$ *set of isomorphism classes of irreducible S-modules.*

Proposition.
$$\bigoplus_{W\in\hat{S}} \mathrm{Hom}_S(W,V)\otimes W \to V, \quad f\otimes w \mapsto f(w)$$
is an S-morphism. If iso, then V semi-simple. If V semi-simple and all irreducible submod of V are finite, then iso.

Definition. *Topological definitions:*

- ***Irreducible*** *M has no non-zero proper closed S-submodules.*
- ***Indecomposable*** *M is not a direct sum of two non-zero closed S-modules.*
- ***Semisimple*** *M is the closure of a direct sum of irreducible S-modules.*

Proposition. *V finite-dimensional $\mathbb{R},\mathbb{C}$-vector space. Let S topological space and $a : S \to \mathrm{End}(V)$. Then*

- *a continuous $\iff$ $S\times V\to V$ continuous*
- *algebraic and topological definitions agree*

Definition.

- *Repr. of top. group G is a topological G-module (π, V_π) with V_π a $(\mathbb{R},\mathbb{C})$-vs, st $\pi : G\to \mathrm{GL}(V_\pi)$ is a group hom.*
- *Repr. of Lie algebra L over a field k is a L-module (ρ, V_ρ) with V_ρ a l-vs for some extension l/k, st $\rho : L\to \mathfrak{gl}(V)$ is a Lie algebra hom.*

Definition (Hilbert-Schmidt inner product)**.** . $A,B\in \mathrm{End}_{\mathbb{C}}(V)$
$$\langle A,B\rangle_{HS} = \mathrm{tr}(A^*B), \qquad \|A\|_{HS} = \sqrt{\langle A,A\rangle_{HS}}$$

Proposition. *The isomorphism*
$$V\otimes V^* \to \mathrm{End}(V), \quad v\otimes f\mapsto T_{v,f} \coloneqq (w\mapsto f(w)v)$$
is an isometry.

We assume $F=\mathbb{R},\mathbb{C}$

Definition (Top. vs)**.** *V vs with topology st $+$ and $\cdot$ continuous.*

Definition. *V is called*

- ***Banach*** *if V complete and comes from a norm $\|-\| : V\to F$.*
- ***Hilbert*** *if Banach and comes from an inner product $\langle -,-\rangle$.*
- ***Fréchet*** *if V complete and comes from a countable family of seminorms $\{p_n\}_{n\geq 1}$.*
- ***Locally Convex*** *if every nbh of 0 contains a convex nbh of 0. ($\iff$ family of seminorms)*
- ***Quasi-Complete*** *if every closed bounded subset is complete.*

Theorem (Hahn-Banach)**.** *V Hausdorff locally convex top. vs. If $W\subseteq V$ subspace and $\lambda : W\to F$, there exist an extension on V.*

Theorem (Closed Graph)**.** *V,W Frechet. $f : V\to W$ linear is continuous $\iff$ graph of f is closed in $V\times W$.*

Theorem (Open mapping theorem)**.** *V,W Frechet. $f : V\to W$ continuous linear surjective $\implies$ f is open.*

Proposition. *Hilbert $\implies$ Banach $\implies$ Fréchet $\implies$ Locally convex*

Definition. *$A : V\to W$ linear btw Banach. A is bounded ($\iff$ continuous) if*
$$\|A\| = \sup_{\|v\|\leq 1} \|Av\| < \infty$$
$\mathrm{Hom}_{cts}(V,W)$ *is Banach with this norm.*

Theorem. *X compact top. space with measure μ, V quasi-complete top. vs. Then $f : X \to V$ continuous $\implies \exists! v \in V$ st for all $\lambda \in \mathrm{Hom}_{cts}(V, F)$*

$$\lambda(v) = \int_X \lambda(f(x)) d\mu(x)$$

Definition (Resolvent set)**.** *V Banach, $A \in \mathrm{End}(V)$. Resolvent set $\rho(A) = \{\lambda \in F : A - \lambda id_V \text{ bijective}\}$, spectrum $\sigma(A) = F \setminus \rho(A)$.*

Proposition. *$\sigma(A)$ is compact.*

Definition. *A called compact if the closure of $A(B)$ is compact where B is the unit ball of V.*

Definition. *For $A \in \mathrm{End}(V)$, there exists A^* st $\langle Av, w\rangle = \langle v, A^* w\rangle$. A self-adjoint if $A = A^*$ and normal if $AA^* = A^* A$.*

Theorem (Spectral theorem)**.** *A normal compact operator on Hilbert space V. Then*

- *$\sigma(A)$ is discrete, 0 is the only possible accumulation point.*
- *$\lambda \in \sigma(A) \setminus \{0\} \implies \lambda$ is an eigenvalue and V_λ fin. dim.*
- *For $\lambda \neq \mu$, then $V_\mu \perp V_\mu$*
- *$V = \hat{\bigoplus}_{\lambda \in \mathbb{C}} V_\lambda$*

Theorem (Stone-Weierstrass)**.** *X compact Hausdorff, $A \subseteq C(X, \mathbb{C})$ subalgebra st*

- *$1 \in A$*
- *A separates points of X ($\forall x \neq y, \exists f \in A$ st $f(x) \neq f(y)$)*
- *A is closed under complex conjugation*

Then A is dense in $C(X, \mathbb{C})$ with respect to the sup. norm.

Theorem (Arzela-Ascoli)**.** *X compact Hausdorff, $F \subseteq C(X, \mathbb{C})$. Then the closure of F is compact $\iff$*

- *F is equicontinuous*
- *F is pointwise bounded*

Theorem. *G top. comp. group. There exist a functional*

$$\mathcal{C}(G, \mathbb{C}) \to \mathbb{C}, \quad f \mapsto \int_G f(x) dx$$

- *If $f(x) \in \mathbb{R}_{>0}$, then $\int_G f(x) dx > 0$*
- *$\int_G f(x) dx = \int_G f(gx) dx = \int_G f(xg) dx$ for all $g \in G$*
- *$\int_G 1 dx = 1$*

Definition. *V representation of G. Then π is called*

- ***Banach representation** if V_π is Banach and π preserves the norm.*
- ***Hilbert representation** if V_π is Hilbert and π preserves the inner product.*
- ***lcc representation** if V_π is Hausdorff, loc. conv. and complete.*

Proposition. *G compact, then $\mathcal{G} \subset L^2(G) \subset L^1(G)$*